

Instance Block - A code block inside a class (not in a method) that runs when an object is created.

(No keyword; just use { }. A class can have multiple blocks; run in order of appearance.)

Object Arrays - Stores multiple objects of a class in a single array.

Call by Value vs Call by Reference

- **Call by Value:** Java passes a copy; original stays unchanged.
- **Call by Reference:** Method can change original (not in Java).

static - not moving or changing

static keyword - makes variables or methods belong to the class, not objects, saving memory.

“static” keyword is used for

- Variables (class-level, shared across instances)
- Methods (called without object creation)
- Blocks (executed once when the class loads)
- Nested Classes (static inner classes)
- **static variable** stored in JVM's **Method Area**.
- **static method**
 - Declared using static.
 - Belongs to the class, not objects.
 - Can be called without creating an object.
 - Can access only static variables and methods.
 - Cannot directly access instance variables or methods.
- **static block**
 - Declared with static { ... }.
 - Runs once when the class is loaded.
 - Used to initialize static variables or one-time setup.

Garbage Collection - Automatically removes unused objects to free memory and prevent memory leaks.

When is Garbage Collector works?

- making reference null

- assigning reference to other

- using anonymous object

Garbage Collection Timing

- **Auto:** JVM removes unreferenced objects.

- **Manual:** Can request, but JVM decides. it's unpredictable.

Enumeration (Enum) - A special type to define a fixed set of constants

Wrapper Class - A class that wraps a primitive type into an object (`int` → `Integer`).

Encapsulation - defined as the wrapping up of data under a single unit.